

Colombian Collegiate Programming League

CCPL 2026

Round 9 -- September 5

Problems

This set contains 10 problems; pages 1 to 17.

(Borrowed from several sources online.)

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A - Monitoring the Amazon

Source file name: `amazon.c`, `amazon.cpp`, `amazon.java`, or `amazon.py`

A network of autonomous, battery-powered, data acquisition stations has been installed to monitor the climate in the region of Amazon. An order-dispatch station can initiate transmission of instructions to the control stations so that they change their current parameters. To avoid overloading the battery, each station (including the order-dispatch station) can only transmit to two other stations. The destinataries of a station are the two closest stations. In case of draw, the first criterion is to chose the westernmost (leftmost on the map), and the second criterion is to chose the southernmost (lowest on the map).

You are commissioned by Amazon State Government to write a program that decides if, given the localization of each station, messages can reach all stations.

Input

The input consists of an integer N , followed by N pairs of integers X_i, Y_i , indicating the localization coordinates of each station. The first pair of coordinates determines the position of the order-dispatch station, while the remaining $N - 1$ pairs are the coordinates of the other stations. The following constraints are imposed: $-20 \leq X_i, Y_i \leq 20$, and $1 \leq N \leq 1000$. The input is terminated with $N = 0$.

The input must be read from standard input.

Output

For each given expression, the output will echo a line with the indicating if all stations can be reached or not (see sample output for the exact format).

The output must be written to standard output.

Sample Input	Sample Output
<pre>4 1 0 0 1 -1 0 0 -1 8 1 0 1 1 0 1 -1 1 -1 0 -1 -1 0 -1 1 -1 6 0 3 0 4 1 3 -1 3 -1 -4 -2 -5 0</pre>	<pre>All stations are reachable. All stations are reachable. There are stations that are unreachable.</pre>

B - Base Equality

Source file name: base.c, base.cpp, base.java, or base.py

Numerous are the moments I as a programmer have been frustrated by the tedious conversions between decimal numbers and hexadecimal ones. Why have we chosen 10 as a base in our everyday numerical presentations, when 16 seems so practically appealing? Obviously because everyone is not the computer geek I am. Maybe some day the world will fully realise the benefits of the hexadecimal system. In the meantime I have to learn to master the base conversions since most of the time numbers do not resemble one another in different bases.

Sometimes peculiar relationships emerge among the different base representations of numbers though. For instance, I noticed just the other day that $1040_{10} \cdot 4 = 104016_{16}$, i.e. $(1 \cdot 10^3 + 0 \cdot 10^2 + 4 \cdot 10^1 + 0 \cdot 10^0) \cdot 4 = (1 \cdot 16^3 + 0 \cdot 16^2 + 4 \cdot 16^1 + 0 \cdot 16^0)$. It made me wonder how often this is the case, that is, the digits of a number in one base, are exactly the same as the digits of a multiple of the number in another base. Formally, let $B_1 < B_2$ be positive integers, and $a_0, a_1, \dots, a_k$ be integers in $[0 \dots B_1 - 1]$. For which a_i 's is there a positive integer c such that

$$\left(\sum_{i=0}^k a_i \cdot B_1^i \right) \cdot c = \left(\sum_{i=0}^k a_i \cdot B_2^i \right).$$

Input

On the first line of input is a positive integer n telling the number of test cases that follow. Each test case is on a line of its own and consists of two integer bases B_1, B_2 , $9 < B_1 < B_2 < 100$, and two integer range elements, r_1 , and r_2 , $0 < r_1 < r_2 < 10000$. Notice that all numbers in the input are given in the base 10.

The input must be read from standard input.

Output

For each test case, there should be one row containing the largest integer i , fulfilling $r_1 < i < r_2$, for which there is a positive integer c such that the digits of i in the base B_1 , are exactly the same as the digits of $i \cdot c$ in the base B_2 . If no such integer i exists, output the text **Non-existent..**

The output must be written to standard output.

Sample Input	Sample Output
4	1040
10 16 1 2000	4240
10 16 1 4999	Non-existent..
10 14 10 9999	9240
11 14 10 9999	

C - Chemical Reaction

Source file name: `chemical.c`, `chemical.cpp`, `chemical.java`, or `chemical.py`

In a chemists lab, there are several types of chemicals in tubes. The chemist wants to mix all these chemicals together, two chemicals at a time. Whenever two chemicals are mixed, some heat is generated and released into the air and the mixed chemical is a known chemical of possibly other type than the original two. The resulting chemical type and the amount of heats emitted can be looked up in the chemical mixture table.

Chemical	1	1	2	2	3	3
	Resulting chemical type	Heats emitted	Resulting chemical type	Heats emitted	Resulting chemical type	Heats emitted
1	1	0	3	-10	3	3000
2	3	-10	2	0	1	-500
3	3	3000	1	-500	3	0

For example, in the above chemical mixture table, there are three types of chemicals: 1, 2, and 3. If you mix chemicals 1 and 3, they produce +3000 units of heat and turn into chemical 3. Sometimes, the heat generated can be negative. For instance, you can mix 2 and 3 and they turn into chemical 1 and in the meantime, cool down the lab by 500 units of heat. Since the chemist lacks funding to buy necessary equipments to protect himself from the heat generated, it is utmost important to find a way to mix all the chemicals together that produces the least total heat, regardless of the final chemical type. For example, suppose the lab has four tubes containing chemicals of types 1, 2, 2, and 3. If the chemicals are mixed in the parenthesize order of $((12)(23))$, it will produce $(-10) + (-500) + (3000) = 2490$ units of heat. However, if the chemicals are mixed in the $(2(1(23)))$ order, it will produce $(-500) + 0 + (-10) = -510$ units of heat, which is also the least total heat possible.

Input

The first line of input file consists of a single number denoting the number of test cases in the file. There is a single line containing a `'/'` character separating two consecutive test cases. The end of the file is marked with a line containing a `'.'`. For each test case, the first line contains an integer number m ($1 \leq m \leq 6$) denoting the number chemical types. Therefore, the chemicals are indexed from 1 up to at most 6. The next $m \times m$ lines give the chemical mixture table entries in the row-major order. Each line contains the new resulting chemical type and the amount of energy emitted. After the table entries is a line with a single number k ($2 \leq k \leq 10$) denoting number of tubes in the lab. The next line then contains k integers in the range of 1 and m , separated by blank spaces, denoting the types of chemicals in those k tubes.

The input must be read from standard input.

Output

For each test case, output the least total heat possible on a single line.

The output must be written to standard output.

Sample Input	Sample Output
2	-510
3	-650
1 0	
3 -10	
3 3000	
3 -10	
2 0	
1 -500	
3 3000	
1 -500	
3 0	
4	
1 2 2 3	
/	
3	
1 0	
3 500	
3 -250	
3 500	
2 0	
1 100	
3 -250	
1 100	
3 0	
6	
1 1 1 2 2 3	
.	

D - Guardian of Decency

Source file name: `decency.c`, `decency.cpp`, `decency.java`, or `decency.py`

Frank N. Stein is a very conservative high-school teacher. He wants to take some of his students on an excursion, but he is afraid that some of them might become couples. While you can never exclude this possibility, he has made some rules that he thinks indicates a low probability two persons will become a couple:

- Their height differs by more than 40 cm.
- They are of the same sex.
- Their preferred music style is different.
- Their favourite sport is the same (they are likely to be fans of different teams and that would result in fighting).

So, for *any two* persons that he brings on the excursion, they must satisfy *at least one* of the requirements above. Help him find the maximum number of persons he can take, given their vital information.

Input

The first line of the input consists of an integer $T \leq 100$ giving the number of test cases. The first line of each test case consists of an integer $N \leq 500$ giving the number of pupils. Next there will be one line for each pupil consisting of four space-separated data items:

- an integer h giving the height in cm;
- a character F for female or M for male;
- a string describing the preferred music style;
- a string with the name of the favourite sport.

No string in the input will contain more than 100 characters, nor will any string contain any whitespace.

The input must be read from standard input.

Output

For each test case in the input there should be one line with an integer giving the maximum number of eligible pupils.

The output must be written to standard output.

Sample Input	Sample Output
2	3
4	7
35 M classicism programming	
0 M baroque skiing	
43 M baroque chess	
30 F baroque soccer	
8	
27 M romance programming	
194 F baroque programming	
67 M baroque ping-pong	
51 M classicism programming	
80 M classicism Paintball	
35 M baroque ping-pong	
39 F romance ping-pong	
110 M romance Paintball	

E - Expanding Rods

Source file name: `expanding.c`, `expanding.cpp`, `expanding.java`, or `expanding.py`

When a thin rod of length L is heated n degrees, it expands to a new length $L' = (1 + n \cdot C) \cdot L$, where C is the coefficient of heat expansion.

When a thin rod is mounted on two solid walls and then heated, it expands and takes the shape of a circular segment, the original rod being the chord of the segment.

Your task is to compute the distance by which the center of the rod is displaced.

Input

The input contains multiple lines. Each line of input contains three non-negative numbers: the initial length of the rod in millimeters, the temperature change in degrees and the coefficient of heat expansion of the material. Input data guarantee that no rod expands by more than one half of its original length. The last line of input contains three negative numbers and it should not be processed.

The input must be read from standard input.

Output

For each line of input, output one line with the displacement of the center of the rod in millimeters with 3 digits of precision.

The output must be written to standard output.

Sample Input	Sample Output
1000 100 0.0001	61.329
15000 10 0.00006	225.020
10 0 0.001	0.000
-1 -1 -1	

F - Fill

Source file name: fill.c, fill.cpp, fill.java, or fill.py

There are three jugs with a volume of a , b and c liters. (a , b , and c are positive integers not greater than 200). The first and the second jug are initially empty, while the third is completely filled with water. It is allowed to pour water from one jug into another until either the first one is empty or the second one is full. This operation can be performed zero, one or more times.

You are to write a program that computes the least total amount of water that needs to be poured; so that at least one of the jugs contains exactly d liters of water (d is a positive integer not greater than 200). If it is not possible to measure d liters this way your program should find a smaller amount of water $d' < d$ which is closest to d and for which d' liters could be produced. When d' is found, your program should compute the least total amount of poured water needed to produce d' liters in at least one of the jugs.

Input

The first line of input contains the number of test cases. In the next T lines, T test cases follow. Each test case is given in one line of input containing four space separated integers — a , b , c and d .

The input must be read from standard input.

Output

The output consists of two integers separated by a single space. The first integer equals the least total amount (the sum of all waters you pour from one jug to another) of poured water. The second integer equals d , if d liters of water could be produced by such transformations, or equals the closest smaller value d' that your program has found.

The output must be written to standard output.

Sample Input	Sample Output
2	2 2
2 3 4 2	9859 62
96 97 199 62	

G - Grapevine

Source file name: `grapevine.c`, `grapevine.cpp`, `grapevine.java`, or `grapevine.py`

In Quadradonia, all rural properties are square, all have the same area, all are perfectly flat and all have the sides aligned to the North-South and West-East axes.

Since properties are flat, the hills in Quadradonia look like a series of huge stairs' steps, with different heights. In a certain mountain, an interesting situation occurs in a rectangular area of $N \times M$ properties. Starting from anywhere within the region, traversing it in the West to East direction, the properties have non-descending heights. Similarly, traversing that region in the North to South direction, starting from anywhere, the properties have also non-descending heights.

A large wine company in Quadradonia wants to rent some properties from that region to grow wine grapes. The company is interested in some special varieties of wine grapes, which are productive only if grown in properties whose heights are within a certain interval. That is, the company is interested in renting properties whose heights are equal to or higher than a given altitude L , and equal to or lower than a given altitude U . To make it easier for harvesting, the rented properties must form a contiguous area. And since everyone in Quadradonia likes squares, the area to be rented must have the shape of a square.

The company has not yet decided which variety of grapes it will grow, and therefore it has a list of queries involving intervals, one for each grape variety. The figure below shows an area of interest of dimensions 4×5 (in number of properties) with examples of areas the company could rent to grow grapes in heights within the intervals given in the picture.

13	21	25	33	34
16	21	33	35	35
16	33	33	45	50
23	51	66	83	93

interval=[20,90]

13	21	25	33	34
16	21	33	35	35
16	33	33	45	50
23	51	66	83	93

interval=[33,35]

13	21	25	33	34
16	21	33	35	35
16	33	33	45	50
23	51	66	83	93

interval=[20,100]

You must write a program that, given the description of the rectangular area of interest in the mountain, and a list of queries containing height intervals, determines, for each query, the largest side, in number of properties, of a contiguous square area with heights within the specified interval.

Input

The input contains several test cases. The first line of a test case contains two integers N and M , separated by a single space, representing respectively the number of properties in the North-South direction ($1 \leq N \leq 500$) and the number of properties in the West-East direction ($1 \leq M \leq 500$) of the region of interest. Each of the next N lines contains M integers $H_{i,j}$, separated by single spaces, indicating the heights of the properties in the region of interest ($0 \leq H_{i,j} \leq 10^5$, for $1 \leq i \leq N$ and $1 \leq j \leq M$; also, $H_{i-1,j} \leq H_{i,j}$ and $H_{i,j-1} \leq H_{i,j}$). The next line contains an integer Q indicating the number of queries ($1 \leq Q \leq 10^4$). Each of the next Q lines describes a query, and contains two integers L and U , separated by a single space, indicating one interval of

heights ($0 \leq L \leq U \leq 10^5$). The heights of properties to be rented must be greater than or equal to L and less than or equal to U .

The last test case is followed by a line containing two zeros separated by a single space.

The input must be read from standard input.

Output

For each test case in the input your program must print $Q + 1$ lines. Each of the first Q lines must contain a single integer, indicating the largest side, in number of properties, of a contiguous square area with heights within the interval specified in the respective input query. The last line to be printed for each test case is used as a separator and must contain a single character '-' (known as hyphen or minus sign).

The output must be written to standard output.

Sample Input	Sample Output
4 5	3
13 21 25 33 34	2
16 21 33 35 35	4
16 33 33 45 50	-
23 51 66 83 93	3
3	1
22 90	1
33 35	0
20 100	-
4 4	
1 7 9 11	
5 8 10 12	
7 10 15 17	
11 19 30 41	
4	
6 20	
7 9	
10 10	
13 14	
0 0	

H - Party at Hali-Bua

Source file name: `hali.c`, `hali.cpp`, `hali.java`, or `hali.py`

Dear Contestant,

I'm going to have a party at my villa at Hali-Bula to celebrate my retirement from BCM. I wish I could invite all my co-workers, but imagine how an employee can enjoy a party when he finds his boss among the guests! So, I decide not to invite both an employee and his/her boss. The organizational hierarchy at BCM is such that nobody has more than one boss, and there is one and only one employee with no boss at all (the Big Boss)! Can I ask you to please write a program to determine the maximum number of guests so that no employee is invited when his/her boss is invited too? I've attached the list of employees and the organizational hierarchy of BCM.

Best,

-Brian Bennett

P.S. I would be very grateful if your program can indicate whether the list of people is uniquely determined if I choose to invite the maximum number of guests with that condition.

Input

The input consists of multiple test cases. Each test case is started with a line containing an integer n ($1 \leq n \leq 200$), the number of BCM employees. The next line contains the name of the Big Boss only. Each of the following $n - 1$ lines contains the name of an employee together with the name of his/her boss. All names are strings of at least one and at most 100 letters and are separated by blanks. The last line of each test case contains a single 0.

The input must be read from standard input.

Output

For each test case, write a single line containing a number indicating the maximum number of guests that can be invited according to the required condition, and a word Yes or No, depending on whether the list of guests is unique in that case.

The output must be written to standard output.

Sample Input	Sample Output
6 Jason Jack Jason Joe Jack Jill Jason John Jack Jim Jill 2 Ming Cho Ming 0	4 Yes 1 No

I - Infected Land

Source file name: infected.c, infected.cpp, infected.java, or infected.py

The earth is under an attack of a deadly virus. Luckily, prompt actions of the Ministry of Health against this emergency successfully confined the spread of the infection within a square grid of areas. Recently, public health specialists found an interesting pattern with regard to the transition of infected areas. At each step in time, every area in the grid changes its infection state according to infection states of its directly (horizontally, vertically, and diagonally) adjacent areas.

- An infected area continues to be infected if it has two or three adjacent infected areas.
- An uninfected area becomes infected if it has exactly three adjacent infected areas.
- An area becomes free of the virus, otherwise.

Your mission is to fight against the virus and disinfect all the areas. The Ministry of Health lets an anti-virus vehicle prototype under your command. The functionality of the vehicle is summarized as follows:

- At the beginning of each time step, you move the vehicle to one of the eight adjacent areas. The vehicle is not allowed to move to an infected area (to protect its operators from the virus). It is not allowed to stay in the same area.
- Following vehicle motion, all the areas, except for the area where the vehicle is in, change their infection states according to the transition rules described above.

Special functionality of the vehicle protects its area from virus infection even if the area is adjacent to exactly three infected areas. Unfortunately, this virus-protection capability of the vehicle does not last. Once the vehicle leaves the area, depending on the infection states of the adjacent areas, the area can be infected.

The area where the vehicle is in, which is uninfected, has the same effect to its adjacent areas as an infected area as far as the transition rules are concerned.

Your mission is to find the length of the shortest sequence(s) of vehicle motion commands that can successfully disinfect all the areas.

Input

The input is a sequence of datasets. The end of the input is indicated by a line containing a single zero. Each dataset is formatted as follows:

n

$a_{11}a_{12} \dots a_{1n}$

$a_{21}a_{22} \dots a_{2n}$

$\dots$

$a_{n1}a_{n2} \dots a_{nn}$

Here, n is the size of the grid. That means that the grid is comprised of $n \times n$ areas. You may assume $1 \leq n \leq 5$. The rest of the dataset consists of n lines of n letters. Each letter a_{ij} specifies the state of the area at the beginning: # for infection, . for free of virus, and @ for the initial location of the vehicle. The only character that can appear in a line is #, ., or @. Among $n \times n$ areas, there exists exactly one area which has @.

The input must be read from standard input.

Output

For each dataset, output the minimum number of time steps that is required to disinfect all the areas. If there exists no motion command sequence that leads to complete disinfection, output -1 . The output should not contain any other extra character.

The output must be written to standard output.

Sample Input	Sample Output
3	0
...	10
.@.	-1
...	3
3	2
.##	1
.#.	6
@##	4
3	
##.	
#..	
@..	
5	
....@	
##...	
#....	
...#.	
##.##	
5	
#...#	
...#.	
#....	
...##	
..@..	
5	
#....	
....	
....	
....	
..@..	
5	
#..#.	
##.##	
.#.#.	
....#	
.#@##	
5	
..##.	
..#..	
#....	
#....	
.#@..	
0	

J - Jaguar King

Source file name: jaguar.c, jaguar.cpp, jaguar.java, or jaguar.py

In a deep forest, a war is going to begin. Like other animals, the jaguars are preparing for this ultimate battle. Though they are mighty strong and lightening fast, they have an extra advantage over other animals. Its their wise and brave king, the Jaguar King.

The king knows that only speed and strength is not enough for winning the war. They have to make a perfect formation. The king has set up a nice formation and placed all the Jaguar Warriors according to that formation. There are N positions for N jaguar warriors (including the king). The king is marked by 1 and the other jaguar warriors are marked by a number from 2 to N . The warriors are placed according to their number.

But then the king realizes that to make the formation perfect and effective, some positions should have stronger jaguars and some should have faster jaguars. Since the strength and speed of all the jaguars are not equal, the king decided to change the positions of some jaguars. The wise king knows the ability of each and every jaguar, so his decision is perfect but the problem is how to change the jaguars.

One of the wise jaguars has given an idea. The idea is simple. All the jaguars will wait for the king's signal, all eyes upon the king. Suppose the king is in the i th position. The king jumps to j th position and when the jaguar at j th position sees the king coming, he immediately jumps to i th position. The king repeats this procedure until they are formatted like the new formation. Now there is another problem. Collision can occur while jumping. So, some wise jaguars have made a jumping scheme so that no collision can occur. The scheme is noted below:

If the king is in the i th position

```
If (i % 4 = 1) The king can jump to position (i+1), (i+3), (i+4), (i-4)
If (i % 4 = 2) The king can jump to position (i+1), (i-1), (i+4), (i-4)
If (i % 4 = 3) The king can jump to position (i+1), (i-1), (i+4), (i-4)
If (i % 4 = 0) The king can jump to position (i-3), (i-1), (i+4), (i-4)
```

Any position is valid if it lies in between 1 and N .

Actually the king can jump much higher between these positions so, no collision can occur. Now you are one of the prisoners (actually they kept you to eat after the war). You have got a chance to get out alive. You know all their ideas and the new formation. If you can tell the king the minimum number of times the king has to jump to gain the new formation, they could be generous and release you.

Input

The input file contains several sets of inputs. The total number of sets will be less than 50. The description of each set is given below:

Each set starts with one integer N ($4 \leq N \leq 40$) which indicates the total number of jaguar warriors. You can assume that N is multiple of 4. The next line will contain N numbers which indicates the final formation of the jaguars. Consecutive numbers will be separated by a single space.

The input will be terminated by the set where $N = 0$. And this set should not be processed.

The input must be read from standard input.

Output

For each set in the input, you should first print the set number starting from 1. And the next line should be the minimum number of times the king has to jump to gain the new formation.

Check the sample input-output for more details. Output should be formatted like the sample output.

Hope you get out alive.

The output must be written to standard output.

Sample Input	Sample Output
4	Set 1:
1 2 3 4	0
4	Set 2:
4 2 3 1	1
8	Set 3:
5 2 3 4 8 6 7 1	2
8	Set 4:
5 2 8 3 6 7 1 4	7
0	